9. Cells and Cell Custers

Content

- Cell Biology - Milestones, The Cell Theory.
- Microscope - Parts, Slide preparation and observation.
- Cell structure – Plasma membrane and cell organelles.
- Plant tissues – Meristematic and permanent tissues.
- Animal tissues -
- Importance of Stem cells.

Questions & Answers

1. The basic building blocks of living things?
   Cells.
2. Branch of science deals with the study of cells?
   Cell Biology (Cytology)
3. Who was the first to observe cells and call them 'cells'?
   Robert Hooke in 1665.



4. Who discovered some microorganisms by observing the water collected from a pond?
   Antonie van Leeuwenhoek in 1674.



5. Who discovered that all plants are composed of cells?
   Matthias Jakob Schleiden in1838.
6. Who discovered that all animals are composed of cells?
   Theodor Schwann in 1839.
7. Iwho introduced the concept that new cells are formed from pre-existing cells?
   Rudolf Virchow in 1855.
8. Who were the scientists who led to the formation of the cell theory?
   MJ Schleiden, Theodor Schwann, Rudolf Virchow.



9. The Cell Theory:
   - All organisms are made up of one or more cells.
   - Cell is the basic unit of life.
   - New cells originate from pre existing cells.
10. What do you mean by the 'resolution of the eye'?
    Human eye can distinguish two spots having a distance of 0.2 mm in adequate light.
    This distance is called resolution of the eye.
    (To distinguish objects less than 0.2 mm apart, lens is required)
11. Different types of microscope?

Simple microscope	- Single lens. - Works under light. - Magnify an object up to 10 times.
Compound microscope	- More than one lens. - Works under light. - Magnify an object up to 1000 times.
Electron microscope	- Electromagnets as lenses. - Electron beams are used instead of light. - Objects are shown in sizes up to 1 million times.







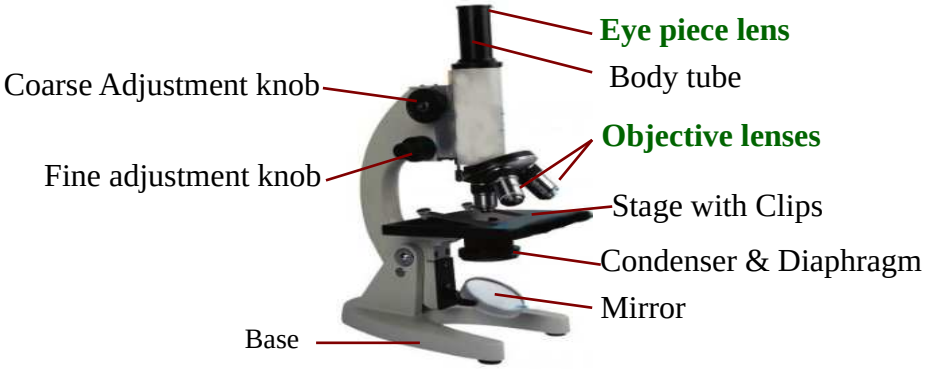
12. Magnification power of a lens?
    The ability of lens to magnify objects.
    (eg:- If the power of eyepiece is 10x and that of objective is 40x, then magnification power will be 10 x 40 = 400x)

13. Who invented electron microscope?
Ernest Raska and Max Knoll in 1934.



14. Which microscope helps to observe cells, viruses and molecular structure in detail?
Electron microscope.
15. Compound microscope: Parts and functions:

Eye piece	The observation object is enlarged again.
Body tube	Holds the eyepiece firmly in place.
Knobes	To adjust the distance to the observation object.
Objective lens	The observation object is enlarged.
Clips	Fixes the slide on the stage.
Condensor lens	Focuses light onto the object of observation.
Condensor, diaphragm	Helps adjust the light to the slide.
Mirror	Reflects light onto the object.



16. How to prepare the observation slide? (eg:- A cross-section of a stem/root):

- Put the material in water to prevent dehydration to the material.
- Take thin sections of the material and keep it in a watch glass, containing water.
- Select and put the thinnest complete sections to a watch glass containing stain.
- Add one or two drops of glycerine to the slide. Place one of the washed sections in it.
- By using a needle, place a cover glass over it without forming air bubbles.
- By using a tissue paper, wipe out the excess glycerine.
- Place this slide on the stage using clips and observe it through the microscope.

17. The substance used to give a specific color to an object to be observed?
Stain. (eg:- Safranine, iosin)
18. What is the purpose of adding glycerin on the slide before putting the observation material?
To prevent dehydration to the observation material.

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19. An example for writing the Observation and Inference in your Practical Record, after observing a cross-section of stem/root using a microscope:

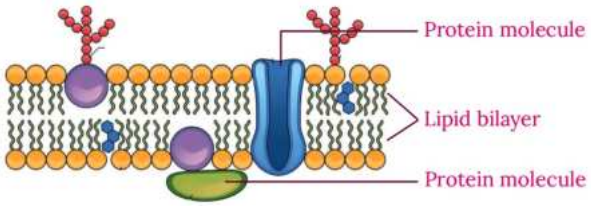
Observation: Within the single-layered sheath, various types of cells are found. Some cells are more colourful. Special types of cells are grouped together to form tissues. In the stem, such tissues are more common on the sides than in the middle. In the leaves, there are both dense and loose cells within the upper and lower layers.

Inference: The unicellular cells are known as the epidermis. There are two epidermises in leaves, one on the upper and one on the lower. In the stem, vascular tissues are seen as rings on the sides. In the leaf, vascular tissues are also seen in the middle.

(*Draw appropriate diagrams too*)

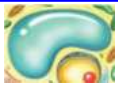
20. The covering of a cell ?
Cell membrane (plasma membrane).
(In plant cell, a cell wall is seen around its cell membrane)

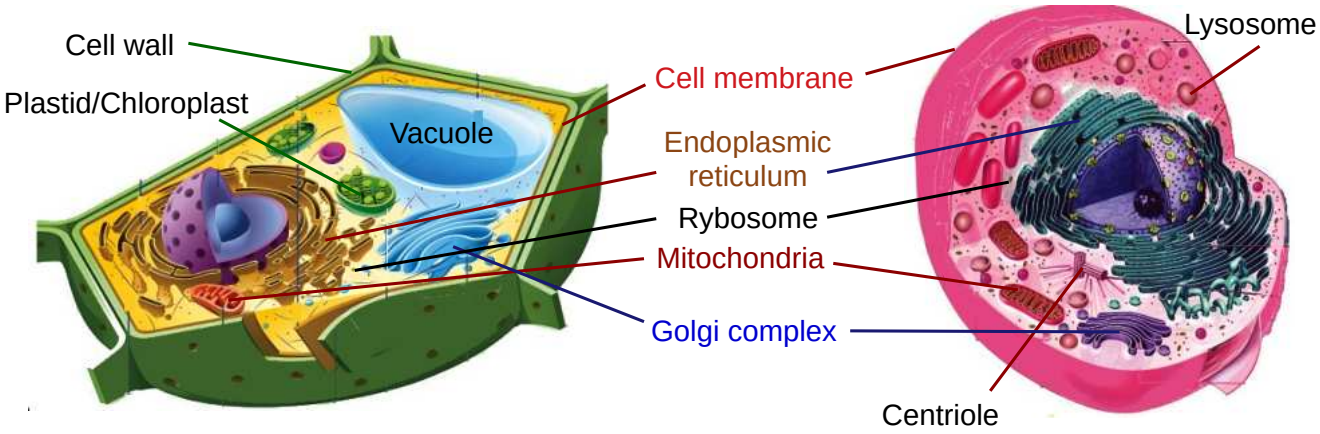
21. Why the plasmamembrane is known as a 'selectively permeable membrane'?
Because the cell membrane allows only certain substances to pass through it.



22. Protoplasm?
All the componets inside the cell membrane is called the protoplasm.
(Cytoplasm + Nucleus = Protoplasm).

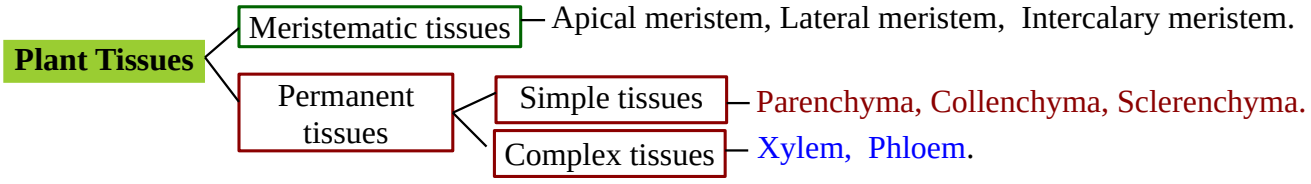
23. Cell organells?
Seen in both the plant and animal cells :-Nucleus, Mitochondrion, Endoplasmic reticulum, Ribosome and Golgi complex.
Seen only in plant cells :- Vacuole and Plastids.
Seen in animal cells :- Centrosome with centrioles and Lysosome.

Parts of cell	Characteristics, Function
Cell wall	- Outer rigid layer, made of cellulose. (seen only in plant cells) - Protects and gives shape to the cell.
Cell membrane (Plasma membrane)	- A thin, flexible layer that surrounds the cell. - It is selectively permeable membrane because it only allows certain substances to pass through. - The membrane through which substances pass into and out of the cell.
Nucleus 	- The centre that controls the cell. - The nucleooplasm contains the chromatin network (that becomes hromosomes) and the nucleolus.
Golgi bodies 	- Cell organelles that appear as stacked membrane layers. - Transports proteins and lipids to various parts wrapped in membranous sacs.
Endoplasmic reticulum 	- Pathway appear as a network of tubules within the cell. - Transports materials needed by the cell.
Mitochondria 	- Energy production centre of the cell. - Sores and distributes the energy obtained from the oxidation of glucose
Plastids 	- Chloroplasts, Chromoplasts, Leucoplasts etc. (seen only in plant cells) - Functions such as synthesizing food, imparting colours, storage, etc..
Ribosome 	- Found in the cytoplasm either freely or attached to endoplasmic reticulum. - Centre for protein synthesis. **Rasheed Odakkal** 9846626323
Vacuole 	- Stores water, nutrients and wastes. (seen in plant cells)
Lysosome	- Destroy foreign substances that enter the cell. (seen in animal cells)
Centrioles	- Help in cell division. (found in centrosomes). (seen in animal cells)



24. During cell division the chromatin networks seen in the controlling centre of cell (nucleus) condense into ----- .
    Chromosomes.
25. The small cell organelles that may seen attached to endoplasmic reticulum?
    Ribosomes.
26. In animal cells, ------ seen inside the centrosome helps in cell division.
    Centrioles.

27. Different types of plastids? Peculiarity? Function of each?
    a. **Chloroplast**
        The green plastids present in leaves and green stems. They contain a green pigment called chlorophyll. It helps to absorb sunlight for synthesising food.
    b. **Chromoplasts**
        These are coloured plastids seen in fruits and flowers. They contain pigments for imparting red, orange and yellow colours, that attract the animals for pollination and seed dispersal.
    c. **Leucoplasts**
        White or colourless plastids found in seeds, roots and stems.
        Leucoplasts store substances like starch,oil and protein.
28. Tissues?
    The group of similar type of cells are known as tissues.
29. Different types of plant tissues?
    Meristematic tissues and Permanent tissues.
    (*Tissues that have formed from meristematic cells and lost their ability to divide are known as permanent tissues*.)



30. Difference between simple and complex plant tissues?
    Simple tissues consist of identical cells.
    Complex tissues contain cells of different sizes and shapes.

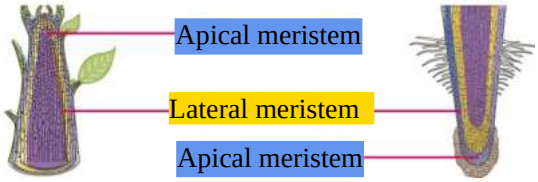
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31. Plant tissues are formed from ----------.
    Meristematic cells.

32. What is the characteristic of a meristem cell compared to a mature plant cell?
    Comparatively the meristematic cells have abundant cytoplasm, large nucleus but thin cell wall and are capable of continuous division. Meristematic are responsible for the growth of plants.

33. Different kinds of meristems?

- **Apical meristem** :- Usually seen at the tips of stem and roots.
- **Lateral meristem** :- Seen in the lateral sides with in the stem and roots. (for girth)
- **Intercalary meristem** :- Seen between the nodes of monocot stem.

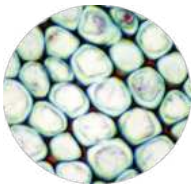
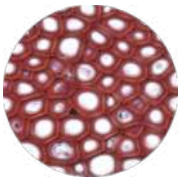




Nodes, where the intercalary meristem seen

34. Permanent simple plant tissues?

Parenchyma	Living, thin-walled cells. Intercellular spaces are present.	Perform photosynthesis and food storage.
Collenchyma	Living cells that have thick walls only in some parts.	Maintains the shape of plant parts.
Sclerenchyma	Dead cells having thick walls on all sides.	Provides strength and support to plant parts.

35. Permanent complex plant tissues?

Xylem :- Conducts water and salt to the leaves from the roots.
Phloem :- Carries food from the leaves to different parts.



Xylem Phloem

36. Animal tissues are formed from ----------.
Stem cells.

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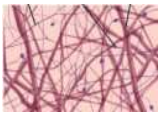
37. What are stem cells? What is their importance?
Stem cells are specialized cells that can develop into different types of cells, such as muscle cells, nerve cells, blood cells, and epithelial cells. They are called the body's master cells because they have the ability to create new types of cells. Stem cells help heal wounds and replace damaged cells. They are even used in healthcare to artificially grow and create body parts.

38. Different types of animal tissues?

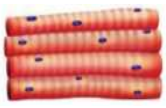
1. **Epithelial tissue**
   Covers and protects the surfaces of the body and internal organs. Helps in the absorption of various substances and also in the production of secretions like mucus.



2. **Connective tissue**
   (Examples:- Blood, bone, cartilage, fibrous tissue)
   Provides support to various parts of the body. Connects various parts together.



3. **Muscle tissue**
   Composed of cells capable of contraction and relaxation. Hence, helps in body movement and locomotion.



4. **Neural tissue**
   Controls and coordinates bodily activities through the transmission of impulses.



Videoclass link: Part 1 :
Part 2 :

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